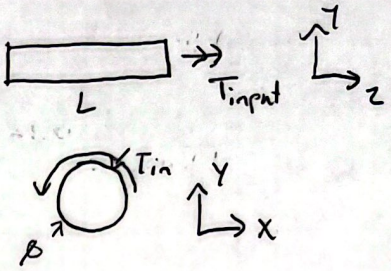
A8

## Input Shaft

- Knowns:
  - $\phi = .125$ in
  - $L = 1$ in
  - $N_{input} = 1100\left(\frac{24}{9}\right) = 2933.33$ rpm
  - $T_{input} = 1.25\left(\frac{7}{24}\right)^2 = .4679$ in lbf
  - $S_y = 4000$ psi
- Unknowns:
  - P
  - T

- FBD:




- Algebraic solutions:

$$P = \frac{D^3 N}{80}$$

$$D_{Torque} = 3\sqrt{\frac{5.1T}{S_y}}$$

$$D_{power} = 3\sqrt{\frac{321000P}{NS_y}}$$

$$T = \frac{T_r}{J} = \frac{T(d/2)}{\frac{\pi d^4}{32}} = \frac{32T(d/2)}{\pi d^4}$$

- Numerical Solutions:

$$P = \frac{.125^3(2933.33)}{80} = .0716 \text{ hp}$$

$$D_T = 3\sqrt{\frac{5.1(.4679)}{4000}} = .0842 \text{ in} \le .125 \text{ in} \checkmark$$

$$D_p = 3\sqrt{\frac{321000(.0716)}{2933.33(4000)}} = .1251 \text{ in} \le .125 \text{ in} \checkmark$$

$$T = \frac{32(.4679)(.125/2)}{\pi(.125)^4} = \boxed{1220.094} \text{ psi} \le 4000 \text{ psi} \checkmark$$

## Output Shaft

- Knowns:
  - $\phi = .25$ in
  - $L = 1$ in
  - $N_{output} = 1100$ rpm
  - $T_{output} = 1.25$ in lbf
  - $S_y = 8500$ psi
- Unknowns:
  - P
  - T
- Algebraic Solutions:

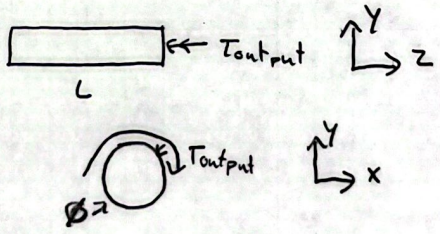
$$P = \frac{D^3 N}{38}$$

$$D_{Torque} = 3\sqrt{\frac{5.1T}{S_y}}$$

$$D_{power} = 3\sqrt{\frac{321000P}{NS_y}}$$

$$T = \frac{32T(d/2)}{\pi d^4}$$

- FBD:




- Numerical Solutions:

$$P = \frac{.25^3(1100)}{38} = .4523 \text{ hp}$$

$$D_T = 3\sqrt{\frac{5.1(1.25)}{8500}} = .0909 \text{ in} \le .25 \text{ in} \checkmark$$

$$D_p = 3\sqrt{\frac{321000(.4523)}{1100(8500)}} = .2494 \text{ in} \le .25 \text{ in} \checkmark$$

$$T = \frac{32(1.25)(.25/2)}{\pi(.25)^4} = \boxed{407.44 \text{ psi}} \le 8500 \text{ psi} \checkmark$$

## Journal Bearing

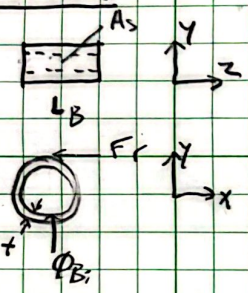
- Knowns:
  - $N_{in} = 2933.33$ rpm
  - $N_{out} = 1100$ rpm
  - $T_{in} = .4689$ in lbf
  - $T_{out} = 1.25$ in lbf
  - $\phi_{in} = .125$ in
  - $\phi_{out} = .25$ in
  - BCS = #7 thru hole
  - BCS = #1,8 thru hole
  - $\phi_{Bi} = .125 + .0007 = .1257$ in
  - $\phi_{Bo} = .25 + .0018 = .2518$ in

- Unknowns:
  - P
  - V
  - PV
  - $A_s$
  - $F_r$
  - t
  - $L_B$

- Material: Graphite dry

### Input Bearing

- FBD:




- Algebraic Solutions:

$$V = N_{in}\left(\frac{\pi \phi_{Bi}}{1}\right)\left(\frac{1}{12}\right)$$

$$P = \frac{F_r}{A_s}$$

$$F_r = \frac{T}{r}$$

$$A_s = L_B \pi(\phi_{Bi}) \quad \text{*Assuming } L_B = \frac{1}{4}L$$

- Numeric Solutions:

$$V = 2933.33\left(\frac{\pi(.1257)}{1}\right)\left(\frac{1}{12}\right) = \boxed{96.53 \text{ ft/min}}$$

$$F_r = \frac{.4689}{\left(\frac{.1257}{2}\right)} = 7.46 \text{ lbf}$$

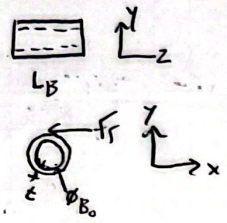
$$A_s = .25(\pi)(.1257) = .0987 \text{ in}^2$$

$$P = \frac{7.46}{.0987} = \boxed{75.58 \text{ psi}}$$

$$PV = 96.53(75.58) = \boxed{7295.74}$$

## Output Bearing

- FBD:




Out Put Material

- Algebraic Solutions:

$$V = N_{out}\left(\frac{\pi \phi_{B_o}}{1}\right)\left(\frac{1}{12}\right)$$

$$P = \frac{F_r}{A_s}$$

$$F_r = \frac{T}{r}$$

$$A_s = L_B \pi (\phi_{B_o})$$

- Numeric Solutions:

$$V = 1100\left(\frac{\pi(.2518)}{1}\right)\left(\frac{1}{12}\right) = \boxed{72.51 \text{ Ft/min}}$$

$$F_r = \frac{1.25}{\left(\frac{.2518}{2}\right)} = 9.93 \text{ lbf}$$

$$A_s = .25(\pi)(.2518) = .1978 \text{ in}^2$$

$$P = \frac{9.93}{.1978} = \boxed{50.2 \text{ psi}}$$

$$PV = 72.51(50.2) = \boxed{3640.002}$$

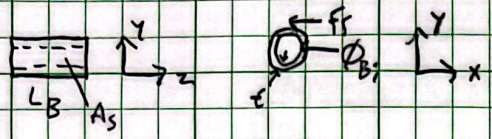
## Bearing Thickness

- Knowns:
  - $P_{in} = 75.58$ psi
  - $P_{out} = 50.2$ psi
  - $V_{in} = 96.53$ ft/min
  - $V_{out} = 72.51$ ft/min
  - $L_B = .25$ in
  - $\phi_{B_i} = .1257$ in
  - $\phi_{B_o} = .2518$ in
  - $F_{r_i} = 7.46$ lbf
  - $F_{r_o} = 9.93$ lbf
  - $A_{s_i} = .0987 \text{ in}^2$
  - $A_{s_o} = .1978 \text{ in}^2$

- Unknowns:
  - $t$

## Input Bearing Thickness

- FBD:




- Algebraic Solutions:

$$P = \frac{F_r}{A} = \frac{F_r}{zt(L_B)} \rightarrow t = \frac{F_r}{zP(L_B)}$$

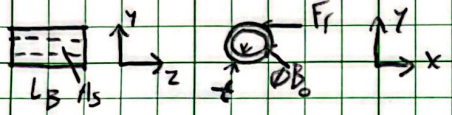
- Numerical Solutions:

$$t = \frac{7.46}{2(25.58)(.25)} = .197 \text{ in}$$

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## Output Bearing Thickness

- FBD:




- Algebraic Solutions:

$$t = \frac{F_r}{zP(L_B)}$$

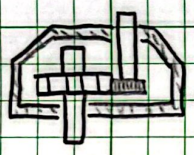
- Numerical Solutions:

$$t = \frac{9.93}{2(50.2)(.25)} = .396 \text{ in}$$

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## Gear Box Design

Top:



Front:



Side:



Fit type: Snap fit

